



Microsoft®
SQL Server®

DATABASE MANAGEMENT SYSTEM

Structured query language

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Introduction to SQL

What is SQL?

SQL stands for Structured Query Language. It is used for storing and managing data in Relational Database Management System (RDBMS). It is a standard language for Relational Database System. It enables a user to create, read, update and delete relational databases and tables. All the RDBMS like MySQL, Informix, Oracle, MSAccess and SQL Server use SQL as their standard database language .

What is DBMS (Database Management System)?

DBMS stands for Database Management System. It is software that allows users to create, manage, and interact with databases efficiently. A DBMS provides an interface between the database and users or application programs, ensuring that data is consistently organized and remains easily accessible.

Key Functions of a DBMS:

1. **Data Storage, Retrieval, and Update:** Stores large volumes of data and allows quick access and modification.
2. **Data Security:** Controls user access and enforces security rules.
3. **Data Integrity:** Maintains accuracy and consistency of data over its lifecycle.
4. **Data Backup and Recovery:** Provides tools for backing up data and restoring it after failures.
5. **Multi-User Access Control:** Allows multiple users to access the database simultaneously without conflicts.

Types of DBMS:

1. **Relational DBMS (RDBMS):** Organizes data into tables (e.g., MySQL, SQL Server).
2. **Hierarchical DBMS:** Organizes data in a tree-like structure.
3. **Network DBMS:** Uses a graph structure for relationships.
4. **Object-Oriented DBMS:** Stores data in objects, as in object-oriented programming

Difference between DBMS (Database Management System) and RDBMS (Relational Database Management System)

Feature	DBMS (Database Management System)	RDBMS (Relational Database Management System)
Data Storage	Stores data as files or collections of data (e.g., hierarchical, network).	Stores data in tables (rows and columns).
Data Relationship	No specific mechanism to define relationships between data.	Enforces relationships between tables using keys (primary/foreign).
Data Integrity	Does not enforce integrity constraints.	Enforces data integrity rules (like entity and referential integrity).
Normalization Support	Usually not supported .	Supports normalization to reduce redundancy.
Query Language	No standard query language (may be proprietary).	Uses SQL (Structured Query Language) .
Example Systems	File systems, XML databases, IMS.	MySQL, SQL Server, Oracle, PostgreSQL.
Multi-user Access	Limited support.	Designed for multi-user environments with concurrency control .
Transactions	Basic or no transaction support.	Supports ACID properties (Atomicity, Consistency, Isolation, Durability).

Summary:

- **DBMS** is a broader concept that includes various types of data management systems.
- **RDBMS** is a specific type of DBMS that organizes data into **related tables**, enforces **rules**, and uses **SQL** for operations.

What is SQL Commands?

SQL commands are instructions. It is used to communicate with the database. It is also used to perform specific tasks, functions, and queries of data.

Types of SQL Commands ?

There are four types of SQL commands:-

- 1.DDL
2. DML
- 3.DCL
4. TCL

1.Data Definition Language (DDL):- DDL changes the structure of the table like creating a table, deleting a table, altering a table, etc.

Here are some commands that come under DDL:

- CREATE
- ALTER
- DROP
- TRUNCATE

A. **CREATE:-** It is used to create a new table in the database.

Syntax: CREATE TABLE TABLE_NAME (COLUMN_NAME
DATA TYPES[,....]);

Example: CREATE TABLE EMPLOYEE(Name VARCHAR2(20), Email
VARCHAR2(100), DOB DATE);

B. **DROP:** It is used to delete both the structure and record stored in the table.

Syntax: DROP TABLE ;

Example: DROP TABLE EMPLOYEE;

C. **ALTER:** It is used to alter the structure of the database. This change could be either to modify the characteristics of an existing attribute or probably to add a new attribute.

Syntax: ALTER TABLE table_name ADD column_name
COLUMN-definition;

ALTER TABLE MODIFY(COLUMN DEFINITION....);

Example: ALTER TABLE STU_DETAILS ADD(ADDRESS VARCHAR2(20));
ALTER TABLE STU_DETAILS MODIFY (NAME VARCHAR2(20));

D. TRUNCATE: It is used to delete all the rows from the table and free the space containing the table.

Syntax: TRUNCATE TABLE table_name;

Example: TRUNCATE TABLE EMPLOYEE;

2.Data Manipulation Language:- DML commands are used to modify the database. It is responsible for all form of CHANGES in the database.

Here are some commands that come under DML:

– INSERT

– UPDATE

– DELETE

A. INSERT: The INSERT statement is a SQL query. It is used to insert data into the row of a table.

Syntax: INSERT INTO TABLE_NAME (col1, col2, col3,... col N)VALUES (value1, value2, value3, valueN);

OR

INSERT INTO TABLE_NAME VALUES (value1, value2, value3, valueN);

Example: INSERT INTO XYZ (Author, Subject) VALUES ("Sonoo", "DBMS");
INSERT:

B. UPDATE: This command is used to update or modify the value of a column in the table.

Syntax: UPDATE table_name SET [column_name1= value1,...column_nameN = valueN] [WHERE CONDITION]

Example: UPDATE students SET User_Name = 'Sonoo' WHERE Student_Id = '3'

C. **DELETE:** This command is used to Delete the value of a column on condition based in the table.

Syntax: DELETE FROM table_name [WHERE CONDITION]

Example: DELETE FROM student where student_id= '3'

3. Data Control Language:- DCL commands are used to GRANT and TAKE BACK authority from any database user.

Here are some commands that come under DCL:

–Grant

–Revoke

A. **GRANT:** It is used to give user access privileges to a database.

Example: GRANT SELECT, UPDATE ON MY_TABLE TO SOME_USER, ANOTHER_USER;

B. **REVOKE:** It is used to take back permissions from the user.

Example: REVOKE SELECT, UPDATE ON MY_TABLE FROM USER1, USER2;

4. Transaction Control Language:- TCL commands can only use with DML commands like INSERT, DELETE and UPDATE only. These operations are automatically committed in the database that's why they cannot be used while creating tables or dropping them.

Here are some commands that come under TCL:

– COMMIT

– ROLLBACK

– SAVEPOINT

A. **COMMIT:-** Commit command is used to save all the transactions to the database.

Syntax: COMMIT;

Example: DELETE FROM CUSTOMERS WHERE AGE = 25; COMMIT;

B. ROLLBACK:-Rollback command is used to undo transactions that have not already been saved to the database.

Syntax: ROLLBACK;

Example: DELETE FROM CUSTOMERS WHERE AGE = 25; ROLLBACK;

C. SAVEPOINT: It is used to roll the transaction back to a certain point without rolling back the entire transaction.

Syntax: SAVEPOINT SAVEPOINT_NAME;

Category	Command	Example	Description
DDL (Definition)	CREATE	CREATE TABLE Members (ID INT, Name VARCHAR(50));	Creates a new table
	ALTER	ALTER TABLE Members ADD Phone VARCHAR(15);	Modifies an existing table
	DROP	DROP TABLE Members;	Deletes a table
DML (Manipulation)	INSERT INTO	INSERT INTO Members (Name) VALUES ('John');	Adds new data
	SELECT	SELECT * FROM Members;	Retrieves data
	UPDATE	UPDATE Members SET Name='Jane' WHERE ID=1;	Modifies existing data
	DELETE	DELETE FROM Members	Removes data
DCL (Control)	GRANT	GRANT SELECT ON Members TO user123;	Gives privileges
	REVOKE	REVOKE SELECT ON Members FROM user123;	Removes privileges
TCL	SAVEPOINT	SAVEPOINT Save1;	Sets a savepoint

(Transaction)			within a transaction
	COMMIT	COMMIT;	Saves changes made in transaction
	ROLLBACK	ROLLBACK;	Reverts changes if an error occurs

SQL OPERATORS:-SQL Operators are special symbols or keywords used in SQL statements to perform operations on data. They are used in SQL queries to filter, compare, and manipulate data in the database.

Types Of SQL Operators

1. Arithmetic Operators
2. Comparison Operators
3. Logical Operators
4. Bitwise Operators
5. Other Special Operators (LIKE, BETWEEN, IN, etc.)

1. Arithmetic Operators

Used for mathematical calculations.

Operator	Description	Example
+	Addition	SELECT 10 + 5;
-	Subtraction	SELECT 10 - 5;
*	Multiplication	SELECT 10 * 5;
/	Division	SELECT 10 / 5;
%	Modulus (remainder)	SELECT 10 % 3;

2. Comparison Operators

Used to compare two values.

Operator	Description	Example
=	Equal	WHERE salary = 50000
!= or <>	Not Equal	WHERE name != 'John'
>	Greater than	WHERE age > 18
<	Less than	WHERE age < 30
>=	Greater than or equal to	WHERE marks >= 40
<=	Less than or equal to	WHERE marks <= 50

3. Logical Operators

Used to combine multiple conditions.

Operator	Description	Example
AND	Returns true if both conditions are true	WHERE age > 18 AND city = 'Delhi'
OR	Returns true if at least one is true	WHERE age < 18 OR city = 'Delhi'
NOT	Reverses the result	WHERE NOT city = 'Delhi'

4. Set Operators

Combine results of queries

Operator	Description
UNION	Combines result sets (removes duplicates)
UNION ALL	Combines result sets (includes duplicates)
INTERSECT	Returns common rows

Other Common SQL Operators

Operator	Description	Example
LIKE	Matches a pattern using wildcards	WHERE name LIKE 'A%'
IN	Checks if a value exists in a list	WHERE dept IN ('HR', 'Finance')
BETWEEN	Checks if a value is within a range	WHERE age BETWEEN 18 AND 25
IS NULL	Checks if a value is NULL	WHERE address IS NULL
EXISTS	Checks if a subquery returns any rows	WHERE EXISTS (SELECT * FROM dept WHERE ...)

Syntax Of Creating Table:-

```
CREATE TABLE table_name (
    column1 datatype [constraints],
    column2 datatype [constraints]
);
```

SQL JOINING:-SQL JOIN means "to combine two or more tables". In SQL JOIN clause is used to combine the records from two or more tables in a database.

1. INNER JOIN

Returns only the rows with matching values in both tables.

Syntax:

```
SELECT A.column1, B.column2 FROM Table AA INNER JOIN Table BB ON
A.common_column = B.common_column;
```

Example:

```
SELECT Employees.Name, Departments.DeptName FROM Employees INNER  
JOIN Departments ON Employees.DeptID = Departments.ID;
```

2. LEFT JOIN (or LEFT OUTER JOIN)

Returns all rows from the left table, and the matched rows from the right table. Unmatched rows will have NULL in right table columns.

Syntax:

```
SELECT A.*, B.* FROM TableA A LEFT JOIN TableB B ON A.common_column  
= B.common_column;
```

3. RIGHT JOIN (or RIGHT OUTER JOIN)

Returns all rows from the right table, and the matched rows from the left table. Unmatched rows from the left side will be NULL.

```
SELECT A.*, B.* FROM TableA A RIGHT JOIN TableB B ON A.common_column =  
B.common_column;
```

4. FULL JOIN (or FULL OUTER JOIN)

Returns all rows from both tables, whether there's a match or not. Where there is no match, NULL is shown.

Syntax:

```
SELECT A.*, B.* FROM TableA A FULL OUTER JOIN TableB B ON  
A.common_column = B.common_column;
```

5. CROSS JOIN

Returns the Cartesian product of two tables — every row from TableA is combined with every row from TableB.

Syntax:

```
SELECT A.*, B.*FROM TableA ACROSS JOIN TableB B;
```

A SELF JOIN is a regular join where a table is joined with itself.

Example:

```
SELECT A.Name AS Employee, B.Name AS ManagerFROM Employees A, Employees B  
WHERE A.ManagerID = B.EmployeeID;
```